

Lesson 1 · Powers, Factorising, Fractions

NCEA Level 2 Mathematics · Algebra

By the end of today you can:

1. simplify any power or root, leaving positive indices
2. factorise completely, whatever the number of terms
3. simplify, combine and solve algebraic fractions

Every step today starts the same way: **factorise first**.

Teaching line, not a question slide.

100 分钟。时间：Do Now 0:00–0:05；核心思想 0:05–0:07；Block 1 指数 0:07–0:20；Block 2 展开和因式分解 0:20–0:35；Block 3 分式 0:35–0:55；Classwork 0:55–1:15；真题演练 1:15–1:35；收尾 1:35–1:40。

发四份纸：Handout 1、Classwork 1、Practice set 1、Homework 1（下课收回 Handout、Classwork、Practice set）。

时间不够时：真题演练只做前 6 页，其余放进下节课开头。

Say it:

algebraic = al-juh-BRAY-ik

factorise = FAK-tuh-rize

indices = IN-dih-seez

end of notes

Do Now

1 Simplify $x^5 \times x^3$.

2 Simplify $\frac{12a^6}{3a^2}$.

3 Expand $(x + 4)(x - 2)$.

4 Factorise $x^2 - 7x + 12$.

5 Work out $\frac{3}{5} - \frac{1}{2}$.

Q1 x^8

Q2 $4a^4$

Q3 $x^2 + 2x - 8$

Q4 $(x - 3)(x - 4)$

Q5 $\frac{1}{10}$

Short solution:

Q2 数字 $12 \div 3 = 4$, 字母 $6 - 2 = 4$

Q4 乘积 12、和 -7 : -3 和 -4

Q5 通分到 10: $\frac{6}{10} - \frac{5}{10}$

Watch for:

Q1 答 x^{15} : 把乘法当成 power of a power

Q3 漏掉中间项, 写成 $x^2 - 8$

Diagnostic:

Q1、Q2 错: Block 1 的 Quick review 多花 2 分钟

Q4 错: Block 2 从 $a = 1$ 的情况讲起, 再讲 $a \neq 1$

Q5 答 $\frac{2}{3}$ (分子减分子、分母减分母): Block 3 的 Quick review 必须做

学生自己改错时题目留在屏幕上

The one idea

$$\frac{3x}{5x} \qquad \frac{x+3}{x+5}$$

You can only cancel _____.

❓ Which one can you cancel? Why not the other one?

Gap factors

Q the first one, $\frac{3x}{5x} = \frac{3}{5}$; the second does not cancel

Short solution:

Q $3x$ 里 x 是 factor (乘上去的), 可以约; $x+3$ 里 x 是 term (加上去的), 不能约

Watch for:

学生说 “ x 约掉剩 $\frac{3}{5}$ ” : 让他代 $x=1$, $\frac{4}{6} \neq \frac{3}{5}$

Ask:

那怎样才能约? (先 factorise, 整个 bracket 相同才约)

Say it:

cancel = KAN-sul

end of notes

Quick review: index laws

1 $x^4 \times x^6 = x$ _____

2 $\frac{y^9}{y^3} = y$ _____

3 $(a^3)^5 = a$ _____

4 $(2x)^3 = \text{_____} x^3$

5 $p^0 = \text{_____}$

Q1 10

Q2 6

Q3 15

Q4 8

Q5 1

Short solution:

Q4 bracket 里每个 factor 都要 power, 数字也要: $2^3 = 8$

Watch for:

Q4 答 2: 数字没有 power (L2), 这是今天最常见的失分

每题 10 秒内答出; 答不出的那一条, 到讲义 Quick review 页再看一遍

Say it:

index = IN-deks

end of notes

Negative indices

Each step down the list divides by 2.

$$2^3 = 8 \quad 2^2 = 4 \quad 2^1 = 2 \quad 2^0 = \underline{\quad} \quad 2^{-1} = \underline{\quad}$$

② Keep going: what is 2^{-2} ?

Gap 1; $\frac{1}{2}$

Q $\frac{1}{4}$

Short solution:

Q 再除以 2: $\frac{1}{2} \div 2 = \frac{1}{4} = \frac{1}{2^2}$

Watch for:

答 -4: 把 negative index 当成负数

Ask:

2^{-3} 呢? ($\frac{1}{8}$)

end of notes

Where does the minus belong?

$$x^{-n} = \frac{1}{x^n}$$

The negative index belongs only to the factor it is written on.

$$3y^{-1} = \frac{3}{\quad}$$

$$(3y)^{-1} = \frac{1}{\quad}$$

❓ In $3y^{-1}$, does the 3 move under the line?

Gap y ; $3y$

Q no, only y moves

Short solution:

Q -1 只写在 y 上，所以只有 y 移到分母； $(3y)^{-1}$ 有括号，整个 $3y$ 都移下去

Watch for:

$$3y^{-1} = \frac{1}{3y} \text{ 是 L1}$$

Ask:

$\frac{1}{x^{-2}}$ 等于什么？ (x^2 : 移到分子, index 变正)

end of notes

Dividing by a negative power

Simplify, leaving positive indices: $\frac{(2y)^3}{4y^{-2}}$

$$\begin{aligned}\frac{(2y)^3}{4y^{-2}} &= \frac{8y^3}{4y^{-2}} \\ &= 2y^{3-(-2)} \\ &= 2y^5\end{aligned}$$

❓ What is $3 - (-2)$, and why write the bracket?

Gap 5

Q 5; the bracket stops you writing $3 - 2$

Short solution:

Q 数字 $\frac{8}{4} = 2$; 字母 index 相减, 减负数等于加: $3 + 2 = 5$

Watch for:

答 $2y^1$: 把 $3 - (-2)$ 算成 $3 - 2$ (L1)

答 $8y^5$ 或 $\frac{8}{4}y^5$ 没约: 数字也要化简

end of notes

Fractional indices

A fraction in the power means a root.

$$x^{\frac{1}{2}} = \sqrt{x}$$

$$x^{\frac{1}{3}} = \sqrt[3]{x}$$

$$8^{\frac{1}{3}} = \underline{\hspace{2cm}}$$

❓ Write $\sqrt{25}$ as a power of 25. What is its value?

Gap 2

$$Q \ 25^{\frac{1}{2}} = 5$$

Short solution:

Q 分母是几就开几次方: $\frac{1}{2}$ 是 square root

Watch for:

把 $8^{\frac{1}{3}}$ 算成 $\frac{8}{3}$: 把 power 当成乘法

Ask:

$27^{\frac{1}{3}}$? (3) $16^{\frac{1}{4}}$? (2)

Say it:

fractional = FRAK-shuh-nul

end of notes

Fractional indices: root first, then power

$$x^{\frac{m}{n}} = \left(\sqrt[n]{x}\right)^m$$

$$8^{\frac{2}{3}} = \left(\sqrt[3]{8}\right)^2 = \underline{\hspace{2cm}}$$

$$16^{-\frac{3}{4}} = \frac{1}{\left(\sqrt[4]{16}\right)^3} = \underline{\hspace{2cm}}$$

❓ Root first or power first: which keeps the numbers small?

Gap 4; $\frac{1}{8}$

Q root first

Short solution:

Q 先开方: $\sqrt[3]{8} = 2$, 再平方是 4; 先平方是 $\sqrt[3]{64}$, 数字变大
负号只决定上下翻, 分母 4 决定开几次方, 分子 3 决定几次方

Watch for:

$16^{-\frac{3}{4}}$ 答成 -8: 负 index 不是负数

end of notes

The root of a power

Root the number. Divide the index by the root.

$$\sqrt{36a^{16}} = 6a^{\frac{16}{2}} = \underline{\hspace{2cm}}$$

$$\sqrt[3]{27b^{12}} = 3b^{\frac{12}{3}} = \underline{\hspace{2cm}}$$

❓ Why is $\sqrt{a^{16}}$ not a^4 ?

Gap $6a^8; 3b^4$

Q a square root halves the index; a^4 takes the square root of the index

Short solution:

Q $\sqrt{a^{16}} = (a^{16})^{\frac{1}{2}} = a^8$; 检查: $a^8 \times a^8 = a^{16}$

Watch for:

数字忘了开方, 写 $36a^8$ (L2)

$\sqrt{a^{16}} = a^4$: 对 index 开方而不是除以 2

end of notes

Spot the error (1)

Simplify $\frac{(3x^2)^3}{9x^{-1}}$, leaving positive indices.

Sam's answer has 2 errors. Find each one, fix it, and say why it loses marks.

Sam's answer

$$\frac{(3x^2)^3}{9x^{-1}} = \frac{3x^6}{9x^{-1}} = \frac{x^6}{3x^{-1}} = \frac{x^5}{3}$$

1. $(3x^2)^3 = 27x^6$, not $3x^6$: the 3 must be cubed too (L2)

2. $x^6 \div x^{-1} = x^{6-(-1)} = x^7$, not x^5 (L1)

Short solution:

$$\frac{27x^6}{9x^{-1}} = 3x^7$$

Watch for:

学生只找到一个错就停：屏幕上写了 2 个，让他继续找

第二个错要说清楚“减负数”： $6 - (-1)$

Ask:

Sam 中间那步 $\frac{3}{9} = \frac{1}{3}$ 对吗？（对，只是用了错的 3）

end of notes

Practice

Simplify, leaving positive indices.

Q1 $\frac{(2a^3)^4}{8a^{-2}}$

Q2 $\sqrt[3]{64m^{15}}$

Q3

$$\left(\frac{9x^4}{x^{10}}\right)^{-\frac{1}{2}}$$

Q1 $2a^{14}$

Q2 $4m^5$

Q3 $\frac{x^3}{3}$

Short solution:

Q1 $\frac{16a^{12}}{8a^{-2}} = 2a^{12-(-2)}$ (u)

Q2 $\sqrt[3]{64} = 4$, $15 \div 3 = 5$ (u)

Q3 先化简括号: $9x^{-6}$; 再 power $-\frac{1}{2}$: $9^{-\frac{1}{2}} = \frac{1}{3}$, $x^{-6 \times (-\frac{1}{2})} = x^3$ (r)

Watch for:

Q1 写 $2a^{10}$: $12 - 2$ (L1)

Q3 先化简括号里面, 数字小很多

end of notes

Expanding three brackets

Expand and simplify $(x + 2)(x - 1)(x + 3)$.

Step 1: expand two of the brackets.

$$\begin{aligned}(x + 2)(x - 1) &= x^2 - x + 2x - 2 \\ &= x^2 + \text{ ______ } x - 2\end{aligned}$$

❓ Which two brackets would you multiply first? Does it matter?

Gap 1

Q any two; the answer is the same

Short solution:

Q 乘法交换律，先乘哪两个都可以；挑数字小的两个更快

Watch for:

$-x + 2x$ 合并成 $-3x$ 或 $3x$: 符号

Say it:

cubic = KYOO-bik

end of notes

Expanding three brackets

Step 2: every term times every term.

$$\begin{aligned} & (x^2 + x - 2)(x + 3) \\ &= x^3 + 3x^2 + x^2 + 3x - 2x - 6 \\ &= \underline{\hspace{2cm}} \end{aligned}$$

❓ How many products should you have before you collect like terms?

Gap $x^3 + 4x^2 + x - 6$

Q six: three terms times two terms

Short solution:

Q 数乘积个数再合并；检查： $x = 1$ 时原式 $3 \times 0 \times 4 = 0$ ，结果 $1 + 4 + 1 - 6 = 0$

Watch for:

只有 4 或 5 个乘积就合并：漏项（L3）

end of notes

Two products worth knowing

$$(a + b)^2 = a^2 + \underline{\hspace{2cm}} + b^2$$

$$(2x - 5)^2 = 4x^2 - 20x + 25$$

❓ Why is $(x + 3)^2$ not $x^2 + 9$?

$$(a - b)(a + b) = \underline{\hspace{2cm}}$$

Gap $2ab$; $a^2 - b^2$

Q it is $(x + 3)(x + 3)$: the two middle products $3x + 3x$ are missing

Short solution:

Q $(x + 3)^2 = x^2 + 6x + 9$; 代 $x = 1$: $16 \neq 10$

Watch for:

$(2x - 5)^2$ 的中间项: $2 \times 2x \times (-5) = -20x$

第二个公式反过来用就是 difference of two squares, 马上要用

end of notes

Factorising: which method?

1. Take out the highest common factor, numbers and letters.
2. Count the terms left in the bracket:
two squares: $a^2 - b^2 = (a - b)(a + b)$
three terms: multiply to c , add to b (or split if $a \neq 1$)
3. Look again: can a bracket go further?

❓ For $2x^2 - 8$, what do you do first?

Q take out the HCF 2: $2(x^2 - 4) = 2(x - 2)(x + 2)$

Short solution:

Q 第一步永远是 HCF; 括号里剩两项, 都是平方, 用 difference of two squares

Watch for:

直接写 $(\sqrt{2}x - \sqrt{8})(\dots)$: 没先提 HCF

停在 $2(x^2 - 4)$: 没有 factorise completely (L4)

end of notes

Step 1: the highest common factor

Factorise $12a^3b^2 - 18a^2b$.

HCF = _____

$12a^3b^2 - 18a^2b = 6a^2b(\text{_____})$

 What is the lowest power of a in both terms?

Gap 6; a^2 ; b ; $2ab - 3$

Q a^2

Short solution:

Q 数字取最大公因数 6; 每个字母取最低次方: a^2 、 b

检查: 展开回去 $6a^2b \times 2ab = 12a^3b^2$

Watch for:

只提 $6a$ 或漏掉 b : 不完全 (L4)

end of notes

When $a \neq 1$: multiply, then split

Factorise $3x^2 + 10x - 8$.

$$a \times c = 3 \times (-8) = -24$$

Two numbers that multiply to -24 and add to 10 : _____ and _____

❓ Which pair: 6 and -4 , 12 and -2 , or 8 and -3 ?

Gap 12 ; -2

Q 12 and -2

Short solution:

Q $12 \times (-2) = -24$, $12 + (-2) = 10$; 其他两组和是 2 和 5

Watch for:

用 $c = -8$ 而不是 $ac = -24$ 找数: $a \neq 1$ 时这是最常见的错

Say it:

coefficient = koh-uh-FISH-unt

end of notes

When $a \neq 1$: multiply, then split

$$3x^2 + 10x - 8 = 3x^2 + 12x - 2x - 8$$

$$= 3x(x + 4) - 2(x + \underline{\quad})$$

$$= (x + 4)(\underline{\hspace{2cm}})$$

❓ Why must the two brackets match before the last step?

Gap 4; $3x - 2$

Q the matching bracket is the common factor you take out

Short solution:

Q $3x(x + 4) - 2(x + 4)$: $(x + 4)$ 是两项的 common factor, 提出来剩 $(3x - 2)$

提 -2 时括号里每个符号都变: $-2x - 8 = -2(x + 4)$

Watch for:

写成 $-2(x - 4)$: 负数提出来符号没变 (L5)

括号不一样, 说明拆分或提公因数错了, 回头检查

end of notes

Factorise completely

$$4x^2 - 25 = (2x - 5)(\underline{\hspace{2cm}})$$

$$3x^3 - 12x = 3x(x^2 - 4) = \underline{\hspace{3cm}}$$

❓ Is $3x(x^2 - 4)$ fully factorised?

Gap $2x + 5$; $3x(x - 2)(x + 2)$

Q no: $x^2 - 4$ is a difference of two squares

Short solution:

Q 提完 HCF 后再看括号：两项且都是平方，还能分

Watch for:

题目写 “Factorise completely” 就要检查每个括号 (L4)

$4x^2 - 25$ 写成 $(4x - 25)(4x + 25)$ 或 $(2x - 25)$ ：数字也要开方 (L2)

end of notes

Spot the error (2)

Factorise completely: **(a)** $6x^2 + 7x - 20$ **(b)** $5x^3 - 45x$

Sam's answer has 2 errors. Find each one, fix it, and say why it loses marks.

Sam's answer

- (a) $6x^2 + 15x - 8x - 20 = 3x(2x + 5) - 4(2x - 5)$
the brackets don't match, so it doesn't factorise
- (b) $5x^3 - 45x = 5x(x^2 - 9)$

1. (a) $-8x - 20 = -4(2x + 5)$, not $-4(2x - 5)$; then $(2x + 5)(3x - 4)$ (L5)

2. (b) $x^2 - 9$ factorises again: $5x(x - 3)(x + 3)$ (L4)

Short solution:

(a) $3x(2x + 5) - 4(2x + 5) = (2x + 5)(3x - 4)$

(b) $5x(x - 3)(x + 3)$

Watch for:

Sam 的结论 “不能分解” 是错的推理推出来的：括号不一样时应该先查符号
展开 $-4(2x - 5)$ 得 $-8x + 20$ ，和原来的 -20 对不上

end of notes

Practice

Q1 Expand and simplify $(x - 3)(x + 2)(2x + 1)$.

Q2 Factorise completely $8x^2 - 18$.

Q3 Factorise $5x^2 - 13x - 6$.

Q1 $2x^3 - x^2 - 13x - 6$

Q2 $2(2x - 3)(2x + 3)$

Q3 $(x - 3)(5x + 2)$

Short solution:

Q1 $(x^2 - x - 6)(2x + 1) = 2x^3 + x^2 - 2x^2 - x - 12x - 6$ (u)

Q2 $2(4x^2 - 9)$, 再用 difference of two squares (u)

Q3 $ac = -30$: -15 和 2 ; $5x(x - 3) + 2(x - 3)$ (u)

Watch for:

Q1 数 6 个乘积 (L3)

Q2 停在 $2(4x^2 - 9)$ (L4)

end of notes

Quick review: number fractions

1 Simplify $\frac{12}{18}$.

2 $\frac{3}{4} \times \frac{8}{9}$

3 $\frac{5}{6} \div \frac{10}{3}$

4 $\frac{5}{6} - \frac{3}{4}$

Q1 $\frac{2}{3}$

Q2 $\frac{2}{3}$

Q3 $\frac{1}{4}$

Q4 $\frac{1}{12}$

Short solution:

Q3 翻转第二个再乘: $\frac{5}{6} \times \frac{3}{10}$

Q4 lowest common denominator 是 12, 不是 24

Watch for:

Q4 用 24 也对, 但最后要约分; 代数分式里用“乘起来”的分母会让化简很难

Say it:

denominator = dih-NOM-ih-nay-ter

end of notes

Simplifying: cancel brackets, not terms

Simplify $\frac{x^2 + 3x}{x^2 - 9}$.

$$= \frac{x(x+3)}{(x-3)(x+3)} = \frac{\underline{\hspace{1cm}}}{\underline{\hspace{1cm}}}$$

❓ Which bracket cancels? Why can't you cancel the x^2 ?

Gap x ; $x - 3$

$Q (x + 3)$ cancels; x^2 is a term, not a factor of the whole top

Short solution:

Q 分子提 x , 分母 difference of two squares; 约掉相同的整个 bracket

Watch for:

约掉 x^2 得 $\frac{3x}{-9} = -\frac{x}{3}$ (L6)

Say it:

numerator = NOO-muh-ray-ter

end of notes

Simplifying when $a \neq 1$

Simplify $\frac{2x^2 + 5x - 3}{4x^2 - 1}$.

$$= \frac{(2x - 1)(x + 3)}{(2x - 1)(\quad)}$$
$$= \underline{\hspace{2cm}}$$

❓ How did $4x^2 - 1$ factorise?

Gap $2x + 1$; $\frac{x + 3}{2x + 1}$

Q difference of two squares: $(2x)^2 - 1^2$

Short solution:

Q 分子: $ac = -6$, 6 和 -1 : $2x^2 + 6x - x - 3 = 2x(x + 3) - (x + 3)$

Watch for:

分子分母都分解后才看哪个 bracket 一样

答案停在没约的形式: 要求 simplified fraction (L8)

end of notes

Multiplying and dividing

Simplify $\frac{x^2 - 4}{3x} \div \frac{x + 2}{6x^2}$.

$$= \frac{(x - 2)(x + 2)}{3x} \times \frac{6x^2}{x + 2}$$
$$= \underline{\hspace{2cm}}$$

❓ What is the first move when you divide by a fraction?

Gap $2x(x - 2)$

Q flip the second fraction and multiply

Short solution:

Q 翻转后: $(x + 2)$ 约掉, $\frac{6x^2}{3x} = 2x$

Watch for:

两个都翻转, 或翻转第一个

答 $2x^2 - 4x$ 也对; factorised 形式更好检查

end of notes

Adding and subtracting

Write $\frac{3}{x+1} + \frac{2}{x-4}$ as a single fraction.

$$= \frac{3(\underline{\hspace{1cm}}) + 2(\underline{\hspace{1cm}})}{(x+1)(x-4)}$$

$$= \frac{5x - 10}{(x+1)(x-4)} = \frac{5(x-2)}{(x+1)(x-4)}$$

❓ What does the top of the first fraction get multiplied by?

Gap $x - 4$; $x + 1$

Q ($x - 4$): the factor its own denominator is missing

Short solution:

Q common denominator 是每个不同 factor 各一次: $(x+1)(x-4)$

最后检查分子能不能提公因数: $5(x-2)$

Watch for:

分子分母分别相加: $\frac{5}{2x-3}$

end of notes

Subtracting: bracket the second top

Write $\frac{4x}{x-2} - \frac{x+3}{x+1}$ as a single fraction.

$$= \frac{4x(x+1) - (x+3)(x-2)}{(x-2)(x+1)}$$

$$= \frac{4x^2 + 4x - (x^2 + x - 6)}{(x-2)(x+1)}$$

$$= \frac{3x^2 + 3x + \underline{\hspace{1cm}}}{(x-2)(x+1)}$$

❓ After the bracket, what does $-(-6)$ become?

Gap 6

Q +6

Short solution:

Q 先把第二个乘积展开放进括号，再整体变号： $-x^2 - x + 6$

最后： $\frac{3(x^2 + x + 2)}{(x-2)(x+1)}$

Watch for:

只变第一项的符号： $-x^2 + x - 6$ (L7)

end of notes

Spot the error (3)

Write $\frac{2x}{x+3} - \frac{x-1}{x-2}$ as a single fraction in its simplest form.

Sam's answer has 2 errors. Find each one, fix it, and say why it loses marks.

Sam's answer

$$\frac{2x(x-2) - (x-1)(x+3)}{(x+3)(x-2)} = \frac{2x^2 - 4x - x^2 + 2x - 3}{x^2 + x - 6} = \frac{x^2 - 2x - 3}{x^2 + x - 6} = \frac{-2x - 3}{x - 6}$$

1. $-(x^2 + 2x - 3) = -x^2 - 2x + 3$: Sam changed only the first sign (L7)

2. The x^2 terms are not factors and cannot cancel (L6)

Short solution:

$$\frac{2x^2 - 4x - x^2 - 2x + 3}{(x+3)(x-2)} = \frac{x^2 - 6x + 3}{(x+3)(x-2)}$$

Watch for:

Sam 第一行是对的：先肯定，再找后面的错

正确答案分子不能分解，所以已经是最简

end of notes

Practice

Q1 Simplify $\frac{x^2 - 5x + 6}{3x - 9}$.

Q2 Write $\frac{2}{x-3} - \frac{x+1}{x^2-9}$ as a single fraction.

Q3 Simplify $\frac{3x^2 - 12}{x^2 + x - 6} \times \frac{x+3}{6x}$.

Q1 $\frac{x-2}{3}$

Q2 $\frac{x+5}{(x-3)(x+3)}$

Q3 $\frac{x+2}{2x}$

Short solution:

Q1 $\frac{(x-2)(x-3)}{3(x-3)}$ (u)

Q2 $x^2 - 9 = (x-3)(x+3)$ 已经包含 $x-3$: $\frac{2(x+3) - (x+1)}{(x-3)(x+3)}$ (r)

Q3 $\frac{3(x-2)(x+2)}{(x+3)(x-2)} \times \frac{x+3}{6x}$ (u)

Watch for:

Q2 用 $(x-3)(x^2-9)$ 作分母: 能做但后面约分很难

Q2 $-(x+1)$ 括号 (L7)

end of notes

Equations with fractions

Solve $\frac{3x+1}{x-2} - 4 = 0$.

Multiply **every term** by $(x-2)$:

$$3x + 1 - 4(\text{_____}) = 0$$

$$3x + 1 - 4x + 8 = 0$$

$$x = \text{_____}$$

❓ What happens to the -4 when you multiply through?

Gap $x - 2$; 9

Q it becomes $-4(x-2)$, which is $-4x+8$

Short solution:

Q 两边每一项都乘 $(x-2)$, 右边 0 还是 0

检查: $x=9$ 时 $\frac{28}{7} - 4 = 0$

Watch for:

写 $3x+1-4=0$: 4 没乘 (L9)

Say it:

equation = ih-KWAY-zhun

end of notes

When an x^2 appears

Solve $\frac{6}{x} + \frac{4}{x+1} = 3$.

Multiply every term by _____:

$$6(x+1) + 4x = 3x(x+1)$$

$$3x^2 - 7x - 6 = 0$$

$$(3x+2)(x-3) = 0$$

❓ Can either answer make a denominator zero?

Gap $x(x+1)$

Q no: $x = 3$ or $x = -\frac{2}{3}$, and only 0 or -1 would

Short solution:

Q $10x + 6 = 3x^2 + 3x$, 移项成 $= 0$ 再分解

检查 $x = 3$: $2 + 1 = 3$

Watch for:

右边的 3 也要乘 $x(x+1)$ (L9)

只写一个解: 二次方程两个都要写

end of notes

Changing the subject

Make x the subject of $y = \frac{2x + 3}{x - 1}$.

$$y(x - 1) = 2x + 3$$

$$xy - y = 2x + 3$$

$$xy - 2x = y + 3$$

$$x(\text{_____}) = y + 3$$

❓ x is in two terms. What single move leaves one x ?

Gap $y - 2$

Q factorise x out

Short solution:

Q 两个含 x 的项放一边，提出 x ，再除： $x = \frac{y + 3}{y - 2}$

Watch for:

写 $x = \frac{y + 3 + 2x}{y}$: x 还在右边 (L10)

end of notes

Practice

Q1 Solve $\frac{2x-1}{x+3} - 1 = 0$.

Q2 Solve $\frac{4}{x-1} - \frac{3}{x} = 1$.

Q3 Make a the subject of $k = \frac{a+4}{3a-2}$.

Q1 $x = 4$

Q2 $x = 3$ or $x = -1$

Q3 $a = \frac{2k+4}{3k-1}$

Short solution:

Q1 $2x - 1 - (x + 3) = 0$ (u)

Q2 乘 $x(x-1)$: $4x - 3(x-1) = x(x-1)$, $x^2 - 2x - 3 = 0$ (r)

Q3 $3ak - 2k = a + 4$, $a(3k - 1) = 2k + 4$ (r)

Watch for:

Q1 -1 乘 $(x+3)$ 后要括号

Q2 两个解都检查: $x = -1$ 时 $-2 + 3 = 1$

end of notes

Classwork 1 · about 20 minutes

- **Part A:** powers, roots and factorising
- **Part B:** fractions: combine, solve, change the subject
- **Part C:** show that ... for every whole number
- **Extension**, if you finish

Teaching line, not a question slide.

答案在 classwork1_answers.pdf (同版面, 红色答案)。

A1 $\frac{5x^7}{2}$; A2 $5p^3$; A3 $3xy(3x - 2y)$; A4 $(x - 3)(2x + 5)$; A5 $\frac{x - 4}{x - 2}$

B1 $\frac{2}{x}$; B2 $x = 1$ or $x = 3$; B3 $t = \frac{1 + 2m}{3 - m}$

C: $(x + 1)^3 - (x - 1)^3 = 6x^2 + 2$, 所以是 6 的倍数加 2

Extension: $x = 3$ or $x = -3$

学生做的时候不要讲; 收卷后先看 Part C 的推理写法

end of notes

(a) Simplify each expression, leaving your answer with positive indices.

(i) $\frac{(3y)^4}{3y^{-1}}$

(ii) $\sqrt[3]{8y^{27}}$

(i) $27y^5$

(ii) $2y^9$

Short solution:

(i) $\frac{81y^4}{3y^{-1}} = 27y^{4-(-1)}$; 评分: Correct expression (u)

(ii) $\sqrt[3]{8} = 2$, $27 \div 3 = 9$ (u)

Watch for:

(i) 写 $27y^3$ (L1) 或 $(3y)^4 = 3y^4$ (L2)

Say it:

expression = ik-SPRESH-un

end of notes

(a) (i) Simplify fully $\sqrt{49y^{36}}$.

(i) $7y^{18}$

Short solution:

(i) $\sqrt{49} = 7$, $36 \div 2 = 18$; 评分: Correct response (u)

Watch for:

$7y^6$: 对 index 开方 (L2)

end of notes

(b) (i) Factorise completely $6x^3y - 15x^2\sqrt{y}$.

(ii) Simplify fully $\frac{6x^2 - x - 12}{3x^2 - 5x - 12}$.

(i) $3x^2\sqrt{y}(2x\sqrt{y} - 5)$

(ii) $\frac{2x - 3}{x - 3}$

Short solution:

(i) $y = \sqrt{y} \times \sqrt{y}$, 所以两项都有 $\sqrt{y}$; 评分: $3x^2(2xy - 5\sqrt{y})$ 只算 u, 完整的才是 r

(ii) $\frac{(2x - 3)(3x + 4)}{(3x + 4)(x - 3)}$ (u)

Watch for:

(i) 没看见 $\sqrt{y}$ 也是公因数 (L4)

(ii) 分子 $ac = -72$: -9 和 8

end of notes

(a) Simplify: $\frac{x^2 - x - 12}{4x + 12}$

(a) $\frac{x - 4}{4}$

Short solution:

(a) $\frac{(x - 4)(x + 3)}{4(x + 3)}$; 评分: Correct simplified fraction (u)

Watch for:

停在没约的形式 (L8)

约成 $x - 1$: $\frac{x - 4}{4}$ 里的 4 不能约 (L6)

end of notes

(b) Write $\frac{5x}{x-3} - \frac{x-4}{x+2}$ as a single fraction in its simplest form.

(b) $\frac{4x^2 + 17x - 12}{(x-3)(x+2)}$

Short solution:

(b) $\frac{5x(x+2) - (x-3)(x-4)}{(x-3)(x+2)}$: 写对这一行是 u，得到正确分式是 r

$$5x^2 + 10x - (x^2 - 7x + 12)$$

Watch for:

$$-(x^2 - 7x + 12) \text{ 每一项变号 (L7)}$$

分子不能分解，所以已经最简

end of notes

(a) Solve $\frac{2x - 3}{x + 4} - 3 = 0$.

(a) $x = -15$

Short solution:

(a) $2x - 3 = 3(x + 4)$, $2x - 3 = 3x + 12$; 评分: Correct solution (u)

Watch for:

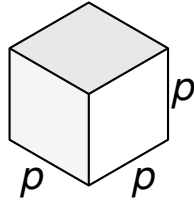
3 没乘 $(x + 4)$ (L9)

end of notes

Practice

Consider a cube with sides of p cm (where $p \neq 0$). The volume of the cube would be $p^3 \text{ cm}^3$, and the surface area of the cube would be $6p^2 \text{ cm}^2$.

Junyang wonders if it is possible to change the dimensions of the cube to make a cuboid that still has the same volume.



Teaching line, not a question slide.

这是 Question Three 的题干，先读题，不答题。下面三页是 (a)(b)(c)

让学生说出 cube 的体积和表面积是怎么来的： $p \times p \times p$ ，6 个面各 p^2

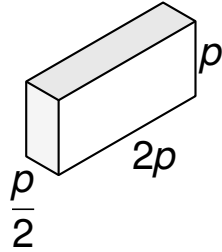
Say it:

cuboid = KYOO-boyd

end of notes

Practice

(a) First, he tries doubling the length, halving the width, and keeping the height as p , as sketched below. Using algebra, find the volume of this cuboid.



(a) p^3

Short solution:

(a) $2p \times \frac{p}{2} \times p = p^3$; 评分: Correct answer (u)

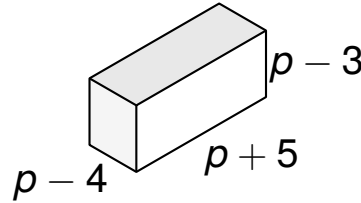
Watch for:

要写出乘法过程 (Using algebra)

end of notes

Practice

(b) The volume of the cuboid below is given by the expression:
 $(p - 4)(p + 5)(p - 3)$.
Expand and simplify this expression.



(b) $p^3 - 2p^2 - 23p + 60$

Short solution:

(b) $(p - 4)(p + 5) = p^2 + p - 20$, 再乘 $(p - 3)$; 评分: Correct simplified expression (u)

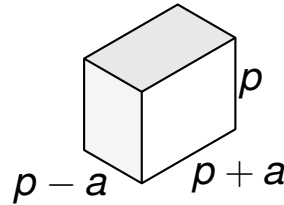
Watch for:

6 个乘积: $p^3 - 3p^2 + p^2 - 3p - 20p + 60$ (L3)

end of notes

(c) Next, Junyang tries adding an amount, a , to the length and then taking off the same amount from the width, keeping the height the same.

Are there any values of a for which the volume of the cuboid will be the same as the volume of the cube?



(c) only $a = 0$, so no: the cuboid never has the same volume unless nothing changes

Short solution:

(c) $p(p + a)(p - a) = p^3 - pa^2$; 令 $= p^3$, $pa^2 = 0$; 因为 $p \neq 0$, $a = 0$

评分: 列到两个体积相等是 u; 有理由的结论是 r

Watch for:

只写 “no” 没有代数过程: 没有分

结论要用题目的语言说: cuboid 和 cube 体积一样只能 $a = 0$

end of notes

Ways to lose marks

- **L1** $3y^{-1} = \frac{3}{y}$; write $6 - (-1)$ in full
- **L2** $(3y)^4 = 81y^4$: power the number too
- **L4** Factorise completely: check every bracket again
- **L6** Cancel whole brackets, never single terms
- **L7** Bracket the second numerator before subtracting
- **L9** Multiply every term by the denominator

Teaching line, not a question slide.

讲义上有完整的 L1–L10; 这里是今天最常见的 6 个

让学生对照自己 Classwork 和 Practice set 的错, 说出是哪一个 L

end of notes

Summary

- Factorise first: cancel, combine and solve on factors, never terms.
- Powers: every factor, number too; a negative index moves the factor across the line.
- Factorising: HCF, count the terms, look again.
- Fractions: one common denominator, bracket the second top, multiply every term.

Teaching line, not a question slide.

让学生用自己的话说一遍第一条，再看作业

end of notes

Homework 1 · about 60 minutes

- **Quick review:** four short items
- **Question One:** powers, roots, factorising
- **Question Two:** algebraic fractions
- **Question Three:** equations and changing the subject

Each question is marked like the real paper: N0 to E8.

Teaching line, not a question slide.

作业是一份三题的小卷子，每题按 u、r、t 给 N0–E8；答案在 hw1_answers.pdf

最后一部分 (e) 是 Excellence 题，先做完 (a)–(d) 再做

下节课开头 5 分钟对答案，按 L 编号说错在哪

end of notes